THANH TRINH-XUAN

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Education

Hanoi University of Science and Technology (HUST) — Hanoi, Vietnam
Bachelor of Engineering in Electrical Engineering — *Aug. 2026 (Expected)*

- **GPA:** 3.2/4.0
- **Research interests:** mathematical programming and distributed algorithms in power systems, peer-to-peer energy markets in multi-microgrids, optimization and operation control in resilience-oriented interconnected smart grids
- **Affiliation:** Member of 100 Renewable Energy Laboratory.
- **Bachelor's Thesis:** Resilience-Oriented Peer-to-Peer Energy Trading in Multi-Microgrids via Network-Constrained Distributed Predictive Control
- **Relevant Coursework:** Power System Analysis, Power System Automation, Power System Protection and Control, Digital System Design, Programming Techniques.

Lam Son High School for the Gifted Students — Thanh Hoa, Vietnam
High School Diploma, Mathematics Specialization — *Jul. 2022*

Research Experience

Undergraduate Researcher — *Sep. 2024 – Present*
Power Grid and Renewable Energy Laboratory, Hanoi, Vietnam
Supervisor: Assoc. Prof. Nguyen Duc Tuyen

Bachelor's Thesis — *2025 – 2026*

- Modeled P-Q-V dynamics using **Second-Order Cone Programming (SOCP)** and integrated load shedding penalty functions to mitigate relaxation inexactness (fictitious currents), ensuring exact grid constraint enforcement.
- Implemented **Analytical Target Cascading (ATC)** to maintain decentralized coordination when extreme faults cause network fragmentation and render centralized control unviable.
- Designed **Model Predictive Control (MPC)** frameworks for dynamic rescheduling to mitigate immediate power imbalances during blackouts that invalidate static day-ahead dispatch.

Project: Distributed Energy Management Frameworks — *2024 – 2025*

- Developed a 3-layer architecture linking **MILP** for discrete unit commitment, **ADMM** for decentralizing tie-lines without exposing private data, and **Nucleolus** for fair **Peer-to-Peer (P2P)** profit sharing.
- Applied **Demand Side Management (DSM)** to optimized local microgrid load scheduling for mitigating peak-load stress and reducing operational costs during islanded mode.
- Implemented an adaptive penalty tuning mechanism based on primal-dual residual balancing to resolve **ADMM** divergence caused by constrained P2P parameters, ensuring rapid convergence.

Lab Contribution & Mentorship — *2024 – Present*

- Architected custom Multi-Agent AI workflows to accelerate personal literature synthesis.
- Mentored junior researchers in academic reading methodologies and algorithmic modeling, driving lab research productivity.

Academic Projects

Machine Learning for EV Load Forecasting *Mar. 2026 – April 2026*

- Engineered a Proxy-Lag Cascade Ensemble architecture utilizing dual XGBoost models (t+24h and t+1) to eliminate phase lag and accurately forecast peak EV load demand.

Industrial Power Supply & Distribution Network Design *Oct. 2025 – Dec. 2025*

- Designed a high/low-voltage distribution network for a large-scale manufacturing plant, optimizing the techno-economic balance between CAPEX and OPEX without compromising system stability.
- Sized transformers, optimized cable cross sections, and performed short circuit calculations to prevent thermal overloads and ensure reliability across all operating conditions.

Electronic & Hardware Prototyping *2023 – 2024*

- Designed and physically prototyped an AC current measurement circuit and a DC-DC Buck converter to ensure stable voltage regulation and high-precision monitoring for hardware testing.
- Simulated layouts using LTspice, Proteus, and MATLAB/Simulink to validate circuit robustness prior to physical fabrication.

Publications & Presentations

[1] **T. X. Thanh (1st Author)**, N. T. Anh, T. X. Hung, N. D. Tuyen, T. T. Son, G. Fujita, "A Decentralized MILP-ADMM Framework for P2P Energy Trading in Interconnected Microgrids," *10th International Conference on Green Energy and Applications (ICGEA)*, Mar. 2026. **[Oral Presentation]**

[2] L. A. Quan, **T. X. Thanh (2nd Author)**, N. T. Anh, P. T. Vinh, T. X. Hung, N. D. Tuyen, T. T. Son, "Microgrid Optimization with MILP-based Demand Side Management," *Journal of Science and Technology (JST)*, vol. 36, Jun. 2026. **[Oral Presentation at Vietnam Student Forum]**

Skills

Optimization: MILP, SOCP, Convex Relaxation

Control & Algorithms: ADMM, ATC, MPC, Game Theory

Programming: Python (Pyomo), C, C++

EE Simulation: Simulink, LTspice, PVsyst, Proteus

Tools & AI: Multi-Agent AI, LaTeX, Git

Languages: English (IELTS 6.0, Target 7.0)

Honors & Awards

2026 Best Presentation Award, 10th Int. Conf. on Green Energy and Applications (ICGEA)

References

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Adjunct Assoc. Prof, Shibaura Institute of Technology, Japan
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